

**Mobile applications.** Applications in Android should be efficient so as not to interfere in the telephony. But this is not the case today. If we try to find the factorial of a number (say, 69) using the scientific calculator in an Android device, it will take almost one second. The performance in this case is seriously affected. We can make use of logical calculations from Vedic Mathematics so as to modify the code and improve the speed of computation.

**Transformer winding machines.** Conventionally, we have transformer winding machines that take data input of parameters, such as primary number of turns and gauge, secondary number of turns and gauge, bobbin and the code. An expert should be made available to gather and feed data. Even in this case, we can program an 8051 microcontroller to automate transformer winding. If we make use of routines based on *sutras*, we can increase the speed of this microcontroller, which is a critical factor in transformer winding.

**Aerofoil controllers.** Aerofoil has a finite number of sensors that gather and send data to the laptop. Here, computations are made and the inclination needed is sent back to activators. The idea is to replace the laptop with a microcontroller that makes use of Vedic Mathematics to improve the speed of computations.

## Roadblocks ahead

**Deciphering the algorithm.** The main difficulty in materialising this concept is in gathering information from the *Vedas*, which is written in



## Sutras and Their Translation

| Sutras                                 | Translation                                |
|----------------------------------------|--------------------------------------------|
| <i>EkadhikinaPurvena</i>               | By one more than the previous one          |
| <i>NikhilamNavatashcaramamDashatah</i> | All from nine and the last from ten        |
| <i>Urdhva-Tiryagbyham</i>              | Vertically and cross-wise                  |
| <i>ParaavartyaYojayet</i>              | Transpose and apply                        |
| <i>ShunyamSaamnyasamuccaye</i>         | When the sum is the same, that sum is zero |
| <i>Anurupyeshunyamanyat</i>            | If one is in ratio, the other is zero      |
| <i>Sankalana-Vyavakalanabhyam</i>      | By addition and by subtraction             |
| <i>Puranapuranaabyham</i>              | By completion or non-completion            |
| <i>Chalana-Kalanabyham</i>             | Differential calculus                      |
| <i>Yaavadunam</i>                      | By deficiency                              |
| <i>Vyasthisamansthi</i>                | Specific and general                       |
| <i>ShesanyankenaCharamena</i>          | Remainders by the last digit               |
| <i>Sopaantyadvayamantyam</i>           | The ultimate and twice the penultimate     |
| <i>Ekanyunenapurvena</i>               | By one less than the previous one          |
| <i>Gunitasamuchyah</i>                 | Product of the sum                         |
| <i>Gunakasamuchyah</i>                 | All multipliers                            |

## Finding the square of a number using sutras (YAVDUNAM TAAVDUNIKRITYA VARGANCHA YOJAYET)

Let us consider the number 98.

$$98^2 = 9604$$

- The nearest power of ten to 98 is 100. Therefore let us take 100 as our base.
- Since 98 is two less than 100, we call two as the deficiency.
- Decrease the given number further by an amount equal to the deficiency, that is, perform  $(98-2) = 96$ . This is the left side of our answer.
- On the right-hand side, put the square of the deficiency, which is square of two = four.
- Append the results from step four and five to get the result. Hence, the answer is 9604.

*Sanskrit.* The challenge is to read and decipher the algorithm encrypted in the scripture. If this is achieved, the engineer can easily code it into microcontrollers.

**Efficient coding.** In order to make the code short and efficient, it should be written in an assembly language—a language that both a microcontroller and programmer understand. Opting for C language for easier coding will consume a lot of program memory. Thus, writing an efficient code is necessarily troublesome for the engineer, but beneficial for users of the product.

## What the future holds

The idea of incorporating Vedic knowledge of Mathematics into microcontrollers was developed by P. Chow Reddy in collaboration with Dr G. Suresh Babu, associate professor

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in electrical and electronics engineering, Chaitanya Bharathi Institute of Technology, Hyderabad. This concept has the potential of revolutionising the embedded industry by improving the performance of microcontrollers by 10-15 folds. Deciphering the algorithm from *Vedas* still remains a major challenge. ●